STA 2002

Lecture 2

Moments of a distribution

Moment-generating function

Central Limit Theorem

Bayes rule

Review of the last lecture

Random variable and distribution function

Expectation and variance

Independence and covariance

Common discrete distribution: binomial, Poisson, geometric, negative binomial, and hypergeometric

Common continuous distribution: uniform, exponential, gamma, normal (Gaussian),χ^2, student t, and F

Moments

The nth *ordinary moment* (or *the moment of the order* n) of X

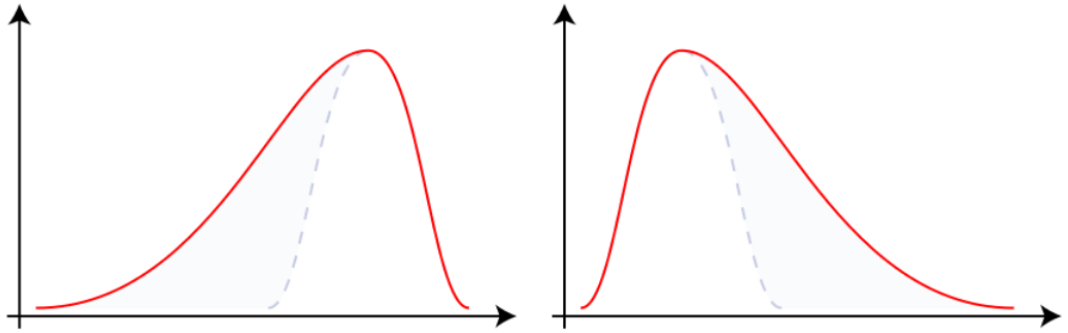
$$m_n = E(X^n)$$

First moment is simply the expectation of X

Second moment is related to variance:

$$Var(X) = E(X^2) - (\mu_X)^2 = E(X^2) - E(X)^2 = m_2 - m_1^2$$

Third moment is related to the skewness of the distribution



Moment-generating function

Definition 3.4.3 Let X be any random variable. Then its *moment-generating function* m_X is defined by $m_X(s) = E(e^{sX})$ at $s \in R^1$.

Example:

Let $X \sim \text{Exponential}(\lambda)$

$$
\begin{aligned}
m_X(s) &= E(e^{sX}) = \int_{-\infty}^{\infty} e^{sx} f_X(x)\,dx = \int_0^{\infty} e^{sx}\lambda e^{-\lambda x}\,dx \\
&= \int_0^{\infty} \lambda\, e^{(s-\lambda)x}\,dx = \frac{\lambda\, e^{(s-\lambda)x}}{s-\lambda}\,\Big|_{x=0}^{x=\infty} = -\frac{\lambda e^{(s-\lambda)0}}{s-\lambda} \\
&= -\frac{\lambda}{s-\lambda} = \lambda(\lambda - s)^{-1}. \blacksquare
\end{aligned}
$$

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)\,.$$

$$F(x) = \int f(x)\,dx.$$

Moment-generating function

Not all random variables have moment-generating functions.

When MGF exists for a random variable, it specifies the distribution.

Binomial $B(n, p)$	$\left(1 - p + pe^t\right)^n$
Poisson $Pois(\lambda)$	$e^{\lambda(e^t - 1)}$
Normal $N(\mu, \sigma^2)$	$e^{t\mu + \frac{1}{2}\sigma^2 t^2}$
Chi-squared X_k^2	$(1 - 2t)^{-\frac{k}{2}}$

Notice we use t here in the place of s in this slide

Moment-generating function

Just in case you are curious:

X~Bin(n, p)

$$\begin{aligned}\mathbf{E}[e^{tX}] &= \sum_{k=0}^{n} e^{tk}\binom{n}{k}p^k(1-p)^{n-k} \\ &= \sum_{k=0}^{n}\binom{n}{k}(pe^t)^k(1-p)^{n-k} = [pe^t+(1-p)]^n\end{aligned}$$

Notice we use t here in the place of s in this slide

X~Poisson(λ)

$$\begin{aligned}M_X(t) &= \sum_{n=0}^{\infty}\frac{\lambda^n e^{-\lambda}}{n!}e^{tn} \\ &= e^{-\lambda}\sum_{n=0}^{\infty}\frac{(\lambda e^t)^n}{n!} \\ &= e^{-\lambda}e^{\lambda e^t} \\ &= e^{\lambda(e^t-1)}\end{aligned}$$

Moment-generating function

Just in case you are curious:

For standard normal,

$$M_z(t) = E(e^{zt}) = \int e^{zt} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

$$\exp\{zt\} \exp\{-\tfrac{1}{2}z^2\} = \exp\{-\tfrac{1}{2}z^2 + zt\}$$
$$= \exp\{-\tfrac{1}{2}(z-t)^2\} \exp\{\tfrac{1}{2}t^2\}.$$

$$M_z(t) = e^{\frac{1}{2}t^2} \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z-t)^2} dz$$
$$= e^{\frac{1}{2}t^2},$$

Notice we use t here in the place of s in this slide

For general normal distribution it is more complex.

Moment-generating function

Suppose that for some $s_0 > 0$, it is true that $m_X(s) < \infty$ whenever $s \in (-s_0, s_0)$. Then

$$\begin{aligned} m_X(0) &= 1 \\ m'_X(0) &= E(X) \\ m''_X(0) &= E(X^2), \end{aligned}$$

In general,

$$m_X^{(k)}(0) = E(X^k),$$

Notice the MGF contains s and model parameters, but NOT X.

X is integrated out.

The derivative is taken against s, and then evaluated at s=0.

$$m_X(s) = E(e^{sX})$$

$$m_X(0) = E(e^{0X}) = E(e^0) = E(1) = 1.$$

$$m'_X(0) = E(X\, e^{0X}) = E(Xe^0) = E(X).$$

$$m''_X(0) = E(X^2\, e^{0X}) = E(X^2 e^0) = E(X^2).$$

Rules of derivatives used here:

Sum rule & constant multiple rule:

$$(af(s)+bg(s))' = (af(s))' + (bg(s))' = af'(s)+bg'(s)$$

Chain rule:

$$(f(g(s)))' = f'(g(s))\, g'(s)$$

Product rule:

$$(f(s)g(s))'=f(s)'\, g(s) + g(s)'\, f(s)$$

Moment-generating function

Let $X \sim \text{Exponential}(\lambda)$

$$m_X(s) = \lambda(\lambda - s)^{-1}$$

$$m'_X(s) = (-1)\lambda(\lambda - s)^{-2}(-1) = \lambda(\lambda - s)^{-2}$$

$$E(X) = m'_X(0) = \lambda(\lambda - 0)^{-2} = \lambda/\lambda^2 = 1/\lambda$$

$$E(X^2) = m''_X(0) = (-2)\lambda(\lambda - 0)^{-3}(-1) = 2\lambda/\lambda^3 = 2/\lambda^2$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = (2/\lambda^2) - (1/\lambda)^2 = 1/\lambda^2$$

$(x^n)' = nx^{n-1}$

Moment-generating function

Poisson distribution

$$m_Z(s) = \exp\left(\lambda(e^s - 1)\right)$$

$$\begin{aligned} E(X) &= m'_X(0) = \lambda e^0 \exp\left(\lambda(e^0 - 1)\right) = \lambda \\ E(X^2) &= m''_X(0) = \lambda e^0 \exp\left(\lambda(e^0 - 1)\right) + \left(\lambda e^0\right)^2 \exp\left(\lambda(e^0 - 1)\right) = \lambda + \lambda^2 \end{aligned}$$

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2 = \lambda + \lambda^2 - \lambda^2 = \lambda$$

$(e^s)' = e^s$

Central Limit Theorem

A large sum of i.i.d. random variables, properly normalized, will always have approximately a *normal* distribution.

Suppose $X_1, X_2, \ldots$ is an i.i.d. sequence of random variables each having finite mean μ and finite variance σ^2. Let $S_n = X_1 + \cdots + X_n$ be the sample sum and $M_n = S_n/n$ be the sample mean. The central limit theorem is concerned with the distribution of the random variable

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\sigma} = \frac{M_n - \mu}{\sigma/\sqrt{n}} = \sqrt{n}\left(\frac{M_n - \mu}{\sigma}\right),$$

As $n \to \infty$, $Z_n \to N(0,1)$

Central Limit Theorem

Example, X follows a Uniform distribution.

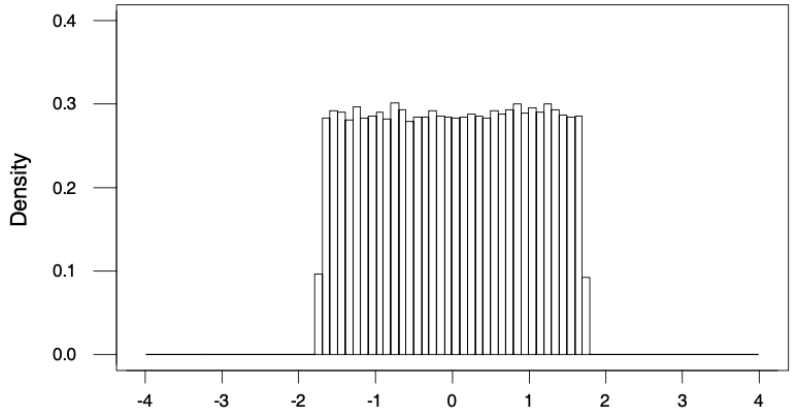


Figure 4.4.2: Density histogram for 10^5 values of Z_1 in Example 4.4.7.

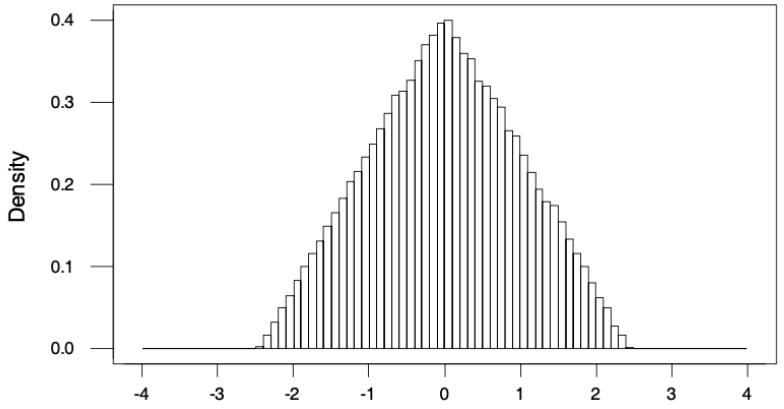


Figure 4.4.3: Density histogram for 10^5 values of Z_2 in Example 4.4.7.

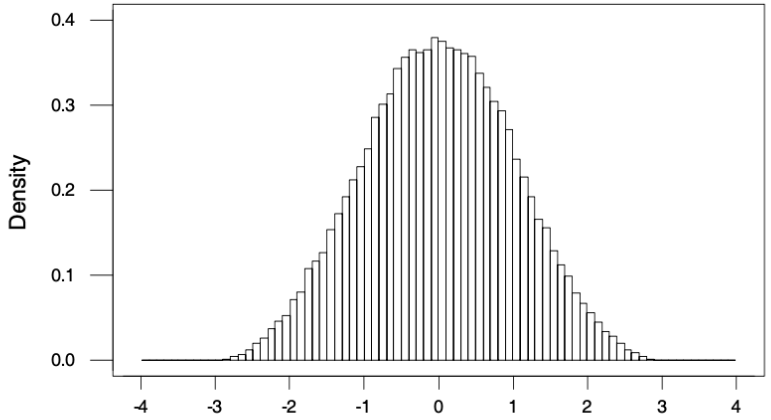


Figure 4.4.4: Density histogram for 10^5 values of Z_3 in Example 4.4.7.

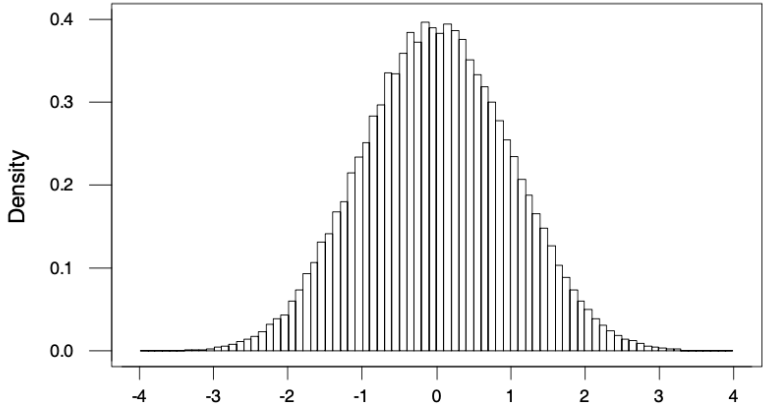


Figure 4.4.5: Density histogram for 10^5 values of Z_{10} in Example 4.4.7.

Central Limit Theorem

$X_1, X_2, \ldots$ are i.i.d. random variables, each with the Poisson(5) distribution.

$$\mu = E(X_i) = 5$$

$$\sigma^2 = \mathrm{Var}(X_i) = 5$$

As $n \to \infty$

$$\sqrt{n}(M_n - 5)/\sqrt{5} \to N(0,1)$$

Central Limit Theorem

$X_1, X_2, \ldots$ are i.i.d. random variables, each with the Bernoulli(θ)

$$E(X_i) = \theta$$

$$\mathrm{Var}(X_i) = \theta\,(1-\theta)$$

The binomial(n, θ) r.v. is the sum of n Bernoulli variables → can use CTL

Let $Y_n = \sum_{i=1}^{n} X_i \sim \mathrm{Bin}(n, \theta)$. As $n \to \infty$,

$$\frac{Y_n - EY_n}{\sqrt{Var(Y_n)}} = \frac{Y_n - n\theta}{\sqrt{n\theta(1-\theta)}} \to N(0,1)$$

$$P\,(Y_n \le y) = P\left(\frac{Y_n - n\theta}{\sqrt{n\theta\,(1-\theta)}} \le \frac{y - n\theta}{\sqrt{n\theta\,(1-\theta)}}\right) \approx \Phi\left(\frac{y - n\theta}{\sqrt{n\theta\,(1-\theta)}}\right)$$

Central Limit Theorem

Suppose that we have a biased coin, where the probability of getting a head on a single toss is $\theta = 0.6$.

We will toss the coin $n = 1000$ times. Calculate the probability of getting at least 550 heads and no more than 625 heads.

$$Y \sim \text{Binomial}(1000, 0.6)$$

$$\begin{aligned} E(Y) &= 1000\,(0.6) = 600, \\ \text{Var}(Y) &= 1000\,(0.6)\,(0.4) = 240. \end{aligned}$$

$$\begin{aligned} P(550 \le Y \le 625) &= P(550 - 0.5 \le Y \le 625 + 0.5) \\ &= P\left(\frac{549.5 - 600}{\sqrt{240}} \le \frac{Y - 600}{\sqrt{240}} \le \frac{625.5 - 600}{\sqrt{240}}\right) \\ &= P\left(-3.2598 \le \frac{Y - 600}{\sqrt{240}} \le 1.646\right) \\ &\approx \Phi(1.65) - \Phi(-3.26) = 0.9505 - 0.0006 = 0.9499. \end{aligned}$$

Central Limit Theorem

Our most intuitive point estimation is justified by CLT.

$X_1, X_2, \cdots, X_n$ are an i.i.d. sequence of random variables with mean μ (unknown)

$$M_n = \frac{X_1 + \cdots + X_n}{n},$$

$$M_n \approx \mu.$$

Consider flipping a sequence of identical coins, each of which has probability θ of coming up heads, where θ is unknown.

Let M_n again be the fraction of the first n coins that are heads.

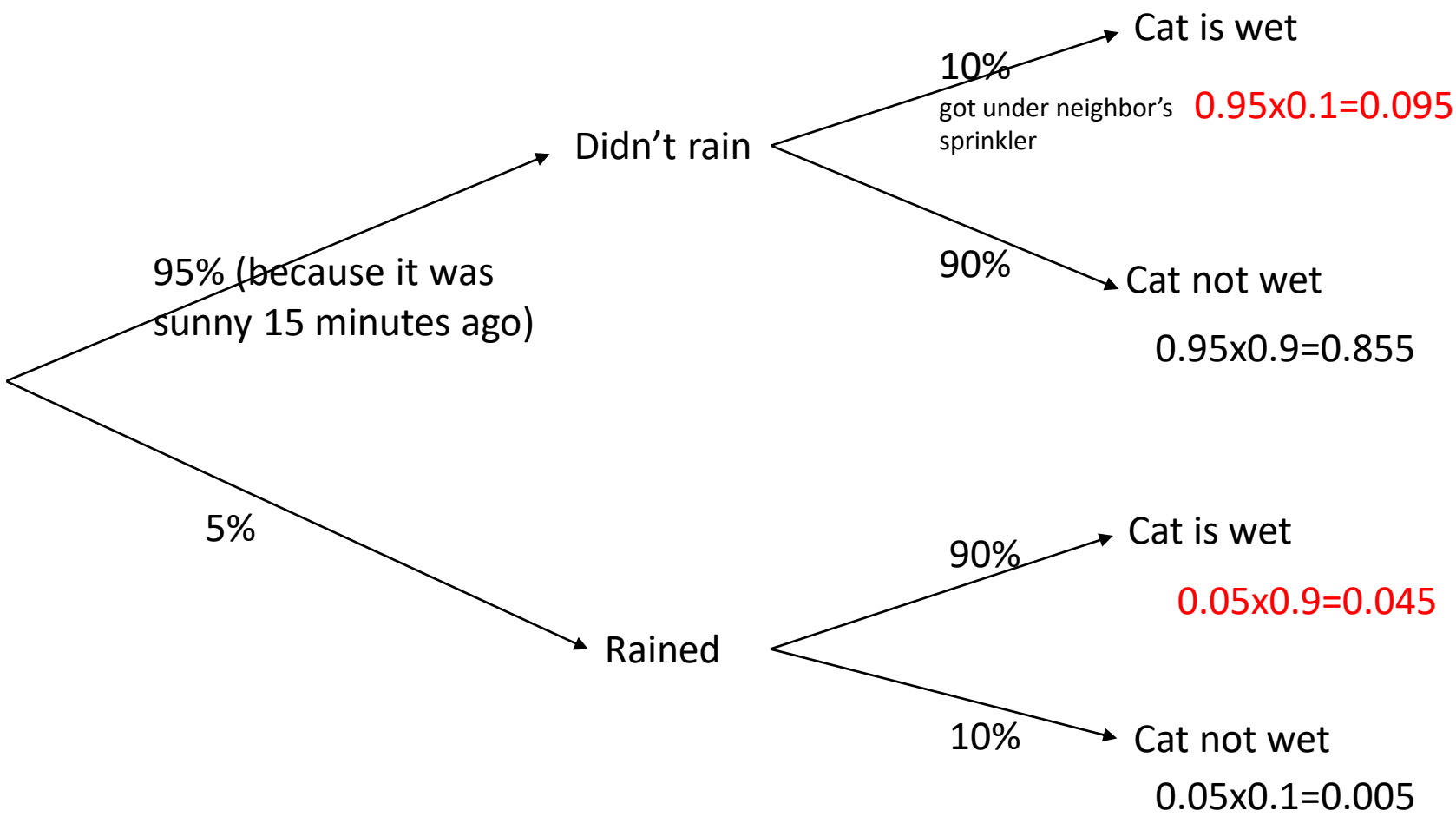
We can use M_n to *estimate* θ.

Bayes' rule

My cat walks in, and I observe he is wet. Without looking outside, what is my guess about the weather?

What I know: 15 minutes ago, it was sunny







Bayes' rule

$$\mathrm{P}(it\ rained \mid cat\ is\ wet)$$

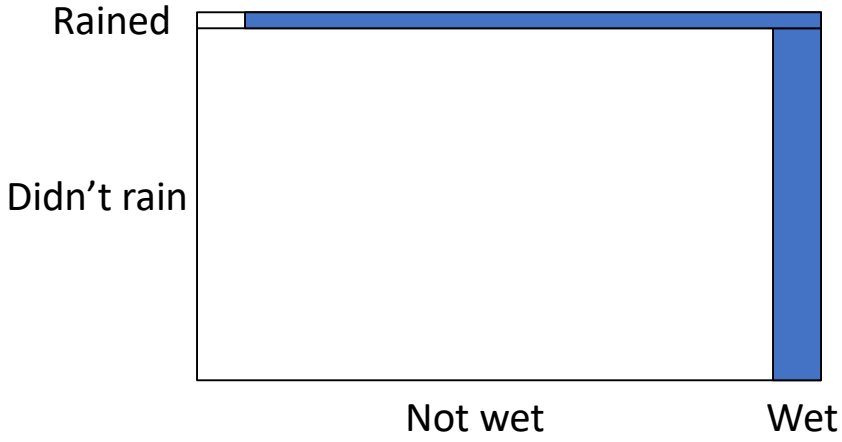
$$=\frac{P(it\ rained\ AND\ cat\ is\ wet)}{P(Cat\ is\ wet)}$$

$$=\frac{P(it\ rained\ AND\ cat\ is\ wet)}{P(it\ rained\ AND\ cat\ is\ wet)+P(it\ didn't\ rain\ AND\ cat\ is\ wet)}$$

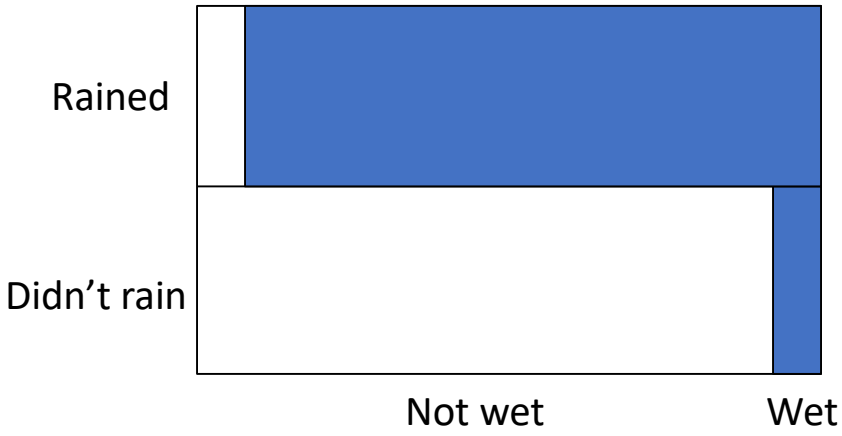
$$=\frac{P(cat\ is\ wet\ |it\ rained)P(it\ rained)}{P(cat\ is\ wet\ |it\ rained)P(it\ rained)+P(cat\ is\ wet\ |it\ didn't\ rain)P(it\ didn't\ rain)}$$

$$=\frac{0.045}{0.045+0.095}$$

$$\approx 32\%$$



However, if I change my prior knowledge about the weather, P(it rained)=0.5, then P(it rained | cat is wet) = 90%



Bayes' rule

BAYES' RULE

Let A and $B_1, \ldots, B_n$ be events where the B_i are disjoint, $\bigcup_{i=1}^{n} B_i = \Omega$, and $P(B_i) > 0$ for all i. Then

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}$$

Ex: T_i: calcification level of coronary arteries
D+: coronary artery disease

i	$P(T_i \mid D+)$	$P(T_i \mid D-)$
0	.42	.96
1	.24	.02
2	.20	.02
3	.15	.00

Bayes' rule

i	$P(T_i \mid D+)$	$P(T_i \mid D-)$
0	.42	.96
1	.24	.02
2	.20	.02
3	.15	.00

For someone 30~39 years old, P(D+)=0.05.
If a person is T_0,

$$P(D+ \mid T_0) = \frac{.42 \times .05}{.42 \times .05 + .96 \times .95} = .02$$

If a person is T_1

$$P(D+ \mid T_1) = \frac{.24 \times .05}{.24 \times .05 + .02 \times .95} = .39$$

For someone 50~59 years old, P(D+)=0.92. Correspondingly,

$$P(D+ \mid T_0) = \frac{.42 \times .92}{.42 \times .92 + .96 \times .08} = .83$$

$$P(D+ \mid T_1) = \frac{.24 \times .92}{.24 \times .92 + .02 \times .08} = .99$$